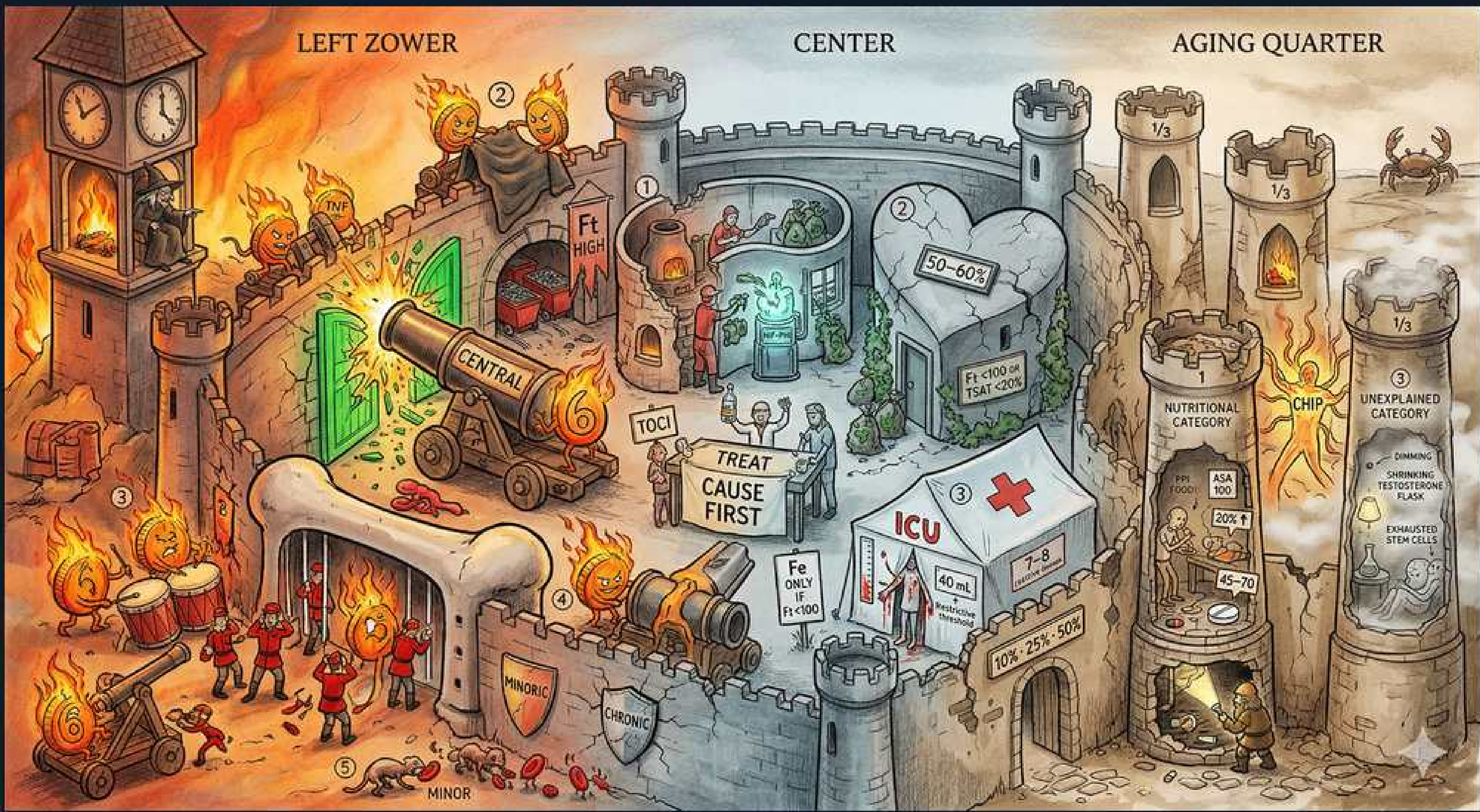
# Iron / Microcytic — anemia of chronic disease & hepcidin (Burning Castle)



### 1. Flaming clock tower (inflammation)

**WHAT YOU SEE**

LEFT tower ablaze, flames everywhere

**WHAT IT ENCODES**

Anemia of chronic disease (anemia of inflammation) is driven by chronic immune activation — infection, malignancy, or autoimmune disease (RA, SLE, IBD). Inflammatory cytokines, chiefly IL-6, stimulate hepatic synthesis of hepcidin, which is the master switch of the whole syndrome. Cytokines also blunt erythropoietin response and shorten RBC survival.

**BOARD TRAP**

WRONG answer: diagnosing ACD in a patient with NO inflammatory/chronic illness. Discriminator — ACD requires an underlying inflammatory source; isolated low iron studies without inflammation point instead to true iron-deficiency anemia (low ferritin, high TIBC).

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### 2. Locked ferritin chests

**WHAT YOU SEE**

Tower 'Ft' chests, flames around stored iron

**WHAT IT ENCODES**

In ACD iron is sequestered in stores, so the pattern is LOW serum iron, LOW TIBC/transferrin, low transferrin saturation, but HIGH or normal FERRITIN (an acute-phase reactant). The body has iron but cannot mobilize it ('functional iron deficiency'). Contrast iron-deficiency anemia: low iron, HIGH TIBC, LOW ferritin.

**BOARD TRAP**

WRONG answer: 'low ferritin confirms ACD.' Discriminator — ferritin is HIGH/normal in pure ACD because inflammation raises it; a genuinely LOW ferritin means true iron deficiency (or coexisting IDA), the single most reliable discriminator on the panel.

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### 3. CENTRAL cannon (hepcidin)

**WHAT YOU SEE**

A big central cannon labelled 'CENTRAL' firing across the castle — the dominating cannon IS hepcidin, the master switch that commands the whole ACD lab picture.

**WHAT IT ENCODES**

Hepcidin is the central hormone of iron regulation: IL-6 drives the liver to produce it, and it works by binding and degrading ferroportin. High hepcidin therefore blocks duodenal iron absorption AND traps iron inside macrophages, lowering serum iron despite full stores. This single mechanism explains the entire ACD lab picture.

**BOARD TRAP**

WRONG answer: 'hepcidin raises serum iron / promotes iron release.' Discriminator — hepcidin LOWERS serum iron by degrading ferroportin; it is the hormone of iron retention, the opposite of what its inflammatory-state elevation might intuitively suggest.

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### 4. FPN / ferroportin gate

**WHAT YOU SEE**

A lower cannon/gate marked 'FPN' — the gate IS ferroportin, the only iron exporter, which hepcidin shuts so iron cannot leave gut or macrophages.

**WHAT IT ENCODES**

Ferroportin is the only cellular iron EXPORTER, found on enterocytes, macrophages, and hepatocytes. Hepcidin internalizes and degrades ferroportin, so iron cannot leave the gut or be recycled from macrophages — producing functional iron trapping with iron-replete but inaccessible stores. This is why oral iron fails in active inflammation.

**BOARD TRAP**

WRONG answer: 'give oral iron — it will correct ACD.' Discriminator — with high hepcidin, ferroportin is gone and gut absorption is blocked, so oral iron is poorly absorbed; if iron is truly needed in inflammation, IV iron bypasses the absorptive defect.

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### 5. TREAT CAUSE FIRST

**WHAT YOU SEE**

A stone banner in the courtyard reading 'TREAT CAUSE FIRST' — the command banner IS the cornerstone of ACD care: fix the underlying inflammation before iron/ESAs.

**WHAT IT ENCODES**

The cornerstone of ACD management is treating the UNDERLYING disease — the anemia is usually mild, normocytic, and resolves as the inflammation is controlled. Iron and ESAs are not first-line and may be unnecessary or harmful if the driver is untreated. Workup also must exclude coexisting true iron deficiency, B12/folate deficiency, and bleeding.

**BOARD TRAP**

WRONG answer: 'start iron and/or EPO as the first intervention in ACD.' Discriminator — first treat the cause; reflexive iron/ESA without addressing inflammation (and without confirming a real iron need) is the classic management error.

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### 6. Mild central anemia

**WHAT YOU SEE**

A heart-shaped doorway labelled '50–60g' at the castle center — the modest heart-gate IS the typically MILD, normocytic anemia of ACD (rarely below 80 g/L).

**WHAT IT ENCODES**

ACD is typically MILD — hemoglobin usually stays in the 90–110 g/L range and rarely falls below 80 g/L — and is NORMOCYTIC-normochromic early, becoming mildly MICROCYTIC only with chronicity. Severe or markedly microcytic anemia should prompt a search for another cause (true IDA, bleeding, thalassemia).

**BOARD TRAP**

WRONG answer: 'severe microcytic anemia is typical of ACD.' Discriminator — pronounced microcytosis and Hb <80 g/L are atypical for pure ACD and suggest concomitant iron deficiency or thalassemia; ACD alone is mild and usually normocytic.

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### 7. ICU / EPO + IV iron

**WHAT YOU SEE**

Red-cross ICU tent, 'Fe only IV/100', '40 g... TCL 25%–50%'

**WHAT IT ENCODES**

For severe or symptomatic ACD (e.g., CKD-related), use erythropoiesis-stimulating agents PLUS adequate iron, and give iron INTRAVENOUSLY because high hepcidin makes oral iron poorly absorbed. Ensure iron repletion before/with ESAs (ESAs need available iron to work) and avoid over-correcting Hb (CKD targets ~100–110 g/L to limit thrombotic risk).

**BOARD TRAP**

WRONG answer: 'oral iron works fine even during active inflammation.' Discriminator — hepcidin blocks enteral absorption, so IV iron is preferred in inflammatory/CKD states; expecting good oral-iron response in active inflammation is the trap.

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### 8. NUTRITIONAL category

**WHAT YOU SEE**

A tower in the aging quarter labelled 'NUTRITIONAL CATEGORY' with elderly figures — it IS one of the three roughly equal buckets of geriatric anemia (iron/B12/folate).

**WHAT IT ENCODES**

In the elderly, anemia is commonly MULTIFACTORIAL and is grouped into three roughly equal buckets: nutritional deficiency (iron, B12, folate), anemia of chronic disease/inflammation (including CKD), and unexplained anemia. Always check iron studies, B12, folate, and renal function in an older anemic patient.

**BOARD TRAP**

WRONG answer: 'assume a single mechanism (e.g., just ACD) in an elderly anemic patient.' Discriminator — geriatric anemia is frequently mixed; missing a coexisting nutritional deficiency (treatable B12/folate/iron) by stopping at one diagnosis is the error.

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### 9. Nutritional flash / deficiency

**WHAT YOU SEE**

A sign reading 'NUTRITIONAL DEFICIENCY FLASH' on the aging-quarter tower — the flash IS combined deficiencies that muddy MCV (high RDW, dimorphic smear).

**WHAT IT ENCODES**

Combined deficiencies muddy the MCV: iron deficiency (microcytic) plus B12/folate deficiency (macrocytic) can yield a NORMAL or mixed MCV with an elevated RDW (anisocytosis) and a dimorphic blood film. Check iron studies, B12, and folate together rather than trusting MCV alone in older patients.

**BOARD TRAP**

WRONG answer: 'a normal MCV rules out both iron and B12/folate deficiency.' Discriminator — opposing deficiencies can cancel out to a normal mean MCV; a high RDW or dimorphic smear unmasks the mixed picture, so MCV alone is unreliable.

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### 10. CHRONIC DISEASE tower

**WHAT YOU SEE**

A tall right-side tower topped by a red crab and marked 'CHR' — the crab tower IS chronic disease/malignancy as a major ACD driver in the elderly (CKD, cancer).

**WHAT IT ENCODES**

Major ACD drivers in the elderly: chronic kidney disease (also adds true EPO deficiency), malignancy (the crab — solid tumors and hematologic cancers), and chronic infection/inflammation. CKD anemia specifically combines reduced erythropoietin production with hepcidin-mediated iron sequestration, which is why it often needs both ESAs and iron.

**BOARD TRAP**

WRONG answer: 'new ACD-type anemia needs no further workup.' Discriminator — unexplained ACD, especially with iron-deficiency features, can be the presenting clue to occult malignancy or GI blood loss; failing to pursue the underlying cause (e.g., colorectal cancer) is the trap.

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### 11. UNEXPLAINED category

**WHAT YOU SEE**

A far-right tower labelled 'UNEXPLAINED CATEGORY' with a slumped, exhausted figure — the worn-out tower IS unexplained geriatric anemia, a diagnosis of exclusion.

**WHAT IT ENCODES**

About a third of elderly anemia is 'UNEXPLAINED' — a diagnosis of exclusion attributed to relative EPO insufficiency, low-grade inflammation, and marrow/stem-cell senescence after ruling out nutritional and chronic-disease causes. It is typically mild and normocytic. Persisting cytopenias or dysplastic features should prompt evaluation for MDS.

**BOARD TRAP**

WRONG answer: 'call anemia unexplained before completing iron/B12/folate/renal and bleeding workup.' Discriminator — unexplained anemia is a diagnosis of EXCLUSION; labeling it prematurely risks missing treatable deficiencies, CKD, or myelodysplastic syndrome.

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### 12. MINOR cannons / IDA overlap

**WHAT YOU SEE**

Small 'MINOR' cannons and tiny red cells at the castle base — the little cannons ARE coexisting TRUE iron deficiency hiding within ACD (use sTfR to detect it).

**WHAT IT ENCODES**

ACD and TRUE iron deficiency frequently COEXIST (e.g., RA or IBD with chronic GI blood loss). When ferritin is ambiguous (the inflammation-inflated 'gray zone,' ~30–100 ng/mL), use the soluble transferrin receptor (sTfR) — HIGH in true iron deficiency, normal in pure ACD — or the sTfR/log-ferritin index to detect the iron-deficient component.

**BOARD TRAP**

WRONG answer: 'a normal/high ferritin excludes iron deficiency in an inflamed patient.' Discriminator — inflammation falsely raises ferritin, so a mid-range ferritin can still hide true IDA; sTfR or the sTfR-ferritin index (or a bone-marrow iron stain) resolves the overlap.

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